JIM DAOU
10/18/26

CPSC 6420 ASSIGNMENT 2

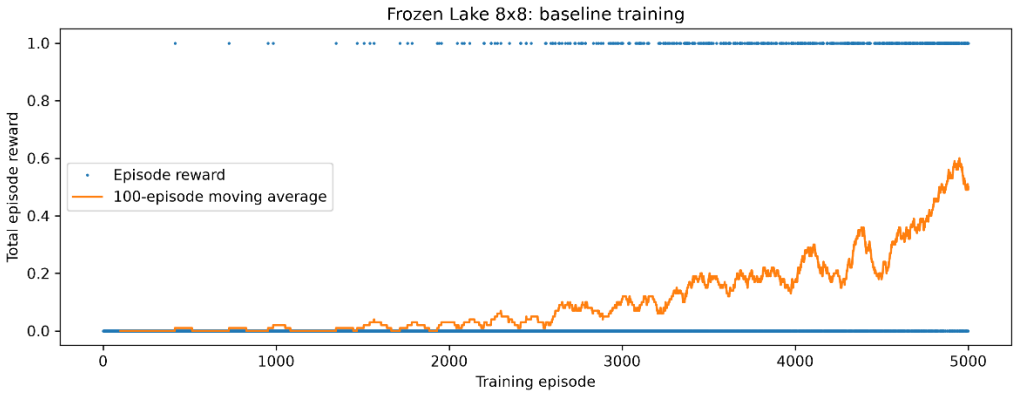
(A) **Environment description**. Describe the environment's goal, state/observation space, action space, transition dynamics, reward structure, and termination conditions. Explain any state discretization or environment settings you use.

a. The environment goal is for the agent to reach the goal(G) from the start(S) while avoiding tiles that have holes(H)
b. Observation space is 64 in the 8x8 map
c. Action space is 4; agent can move up, down, right, and left
d. Transition dynamics of frozen lake depend on if we have defined 'is_slippery'. With it set to 'True', the default behavior causes the agent to only have a 1/3 chance of moving in its intended direction and 1/3 chance of moving in either direction perpendicular to its intended direction. With it set to 'False', the agent always moves in its intended direction.
e. Reward structure for the agent is +1 for reaching the goal and 0 for everything else, including moving to a tile with a hole
f. Termination happens when the agent moves to a tile with a hole or reaches the goal tile. Episode is truncated at 200 steps for the 8x8 environment map.
g. No further state discretization settings or environment settings were used
h. The observations stated here were gathered using 'inspect_frozen_lake.py' and official Frozen Lake documentation

(B) 'agent.py' was created and features an agent that uses Q-Learning

(C) Baseline configuration chosen has epsilon = 1, gamma = 1, alpha = 0.1, and 5000 episodes

a. Epsilon decays at a rate of ({number of episodes} – {number of episodes observed}) / {number of episodes). This can be considered our exploration schedule.
b. This is the plot of a baseline training session. We observed a rolling average that increased overall, but was rough and had lots of ups and downs



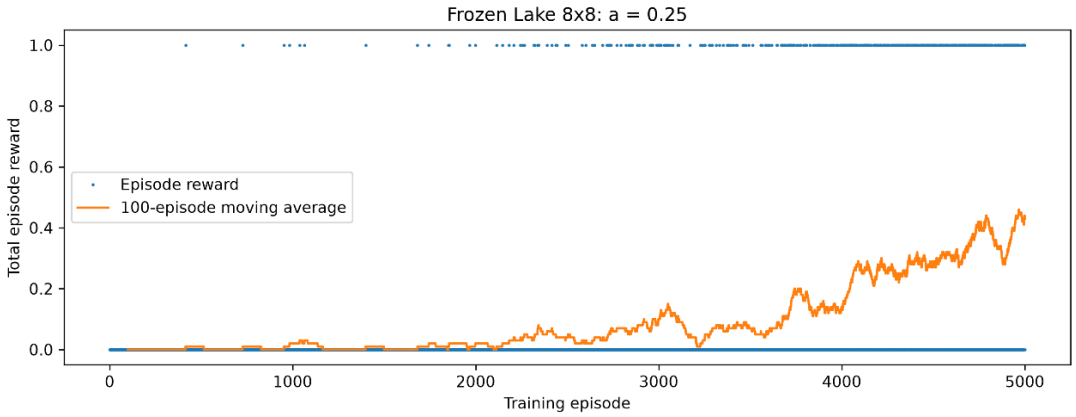
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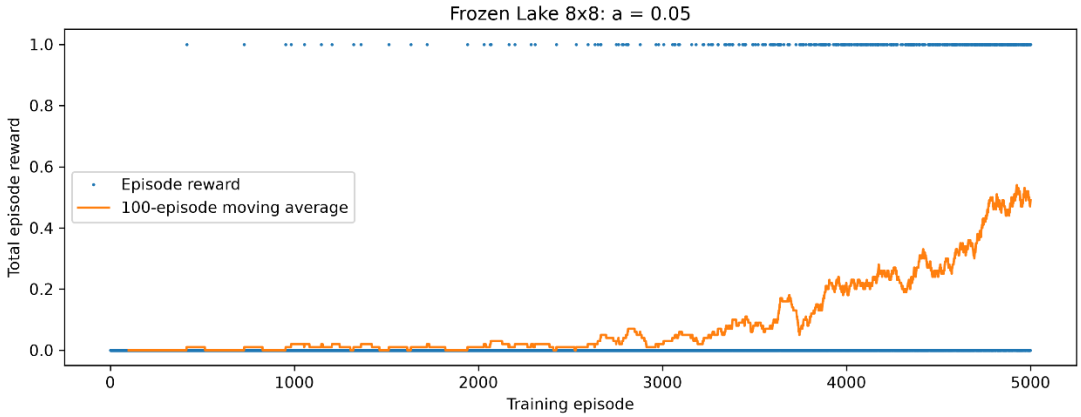
c. This behavior could be attributed to a suboptimal learning rate or the randomness of Frozen Lake's 'is_slippery' feature

(D) Learning rate tests were conducted with alpha 0.25, 0.05, and 0.01. Baseline was used for all other variables.

a. At alpha 0.25, we notice a similar learning speed to the baseline; however, there is less stability and slightly worse overall performance. Higher learning rate did not improve the agent.



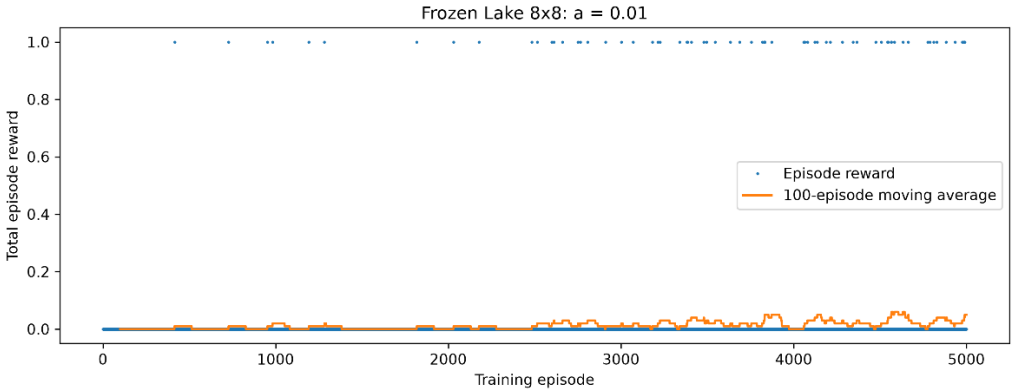
b. At alpha 0.05, there is a noticeably slower learning rate compared to baseline, but stability is slightly smoother. Still, overall performance is about the same.



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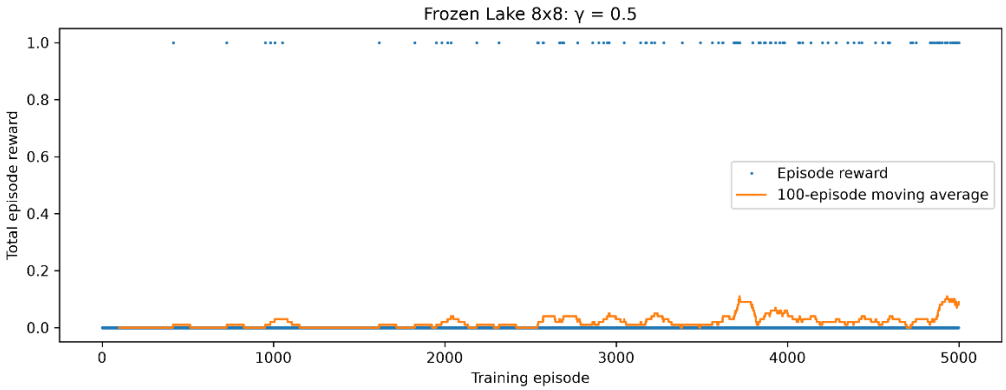
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c. At alpha 0.01, the agent fails to meaningfully learn. There is nearly no observed learning rate, and the overall performance reflects this. An alpha this low likely hinders the agent from meaningfully incorporating new observations.



(E) Discount factor tests were conducted with gamma 0.5, 0.9, and 0.99. Baseline was used for all other variables.

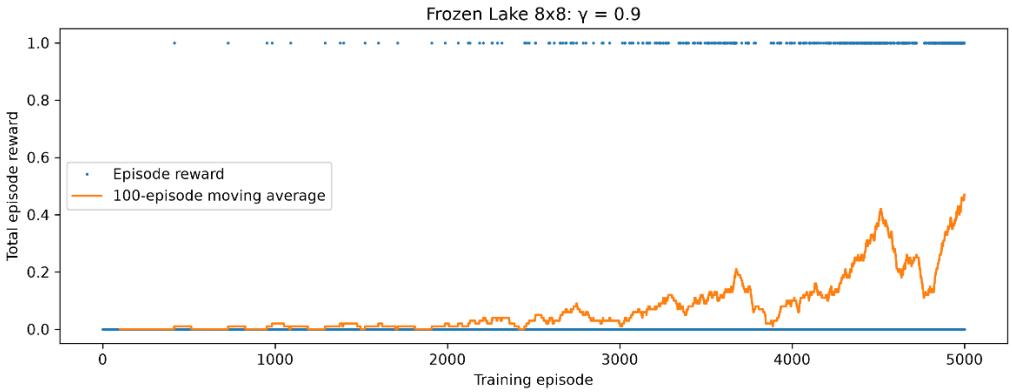
a. At gamma 0.5, there is very slow growth, if any at all. Heavy discounts for delayed rewards do not help this agent, likely due to there being only 1 reward and that reward only appearing after many steps.



b. At gamma 0.9, there initially is accelerated growth to maximum performance, but then there is a steep drop in learning rate followed by a steep increase. This is unusual behavior as a positively affecting gamma should only increase learning rate while a negatively affecting gamma should only lower learning rate. Since the peak of average reward over 100 episodes is achieved sooner, this gamma could be considered to positively affect the agent.

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c. At gamma 0.99, learning rate has the most confident positive trend, despite subpar stability. Still, compared to the baseline, this agent starts learning noticeably around episode 2000, while baseline starts learning noticeably around episode 1500. Further, overall performance for this agent is still slightly worse compared to baseline. Overall, discounts on delayed rewards do not seem to help the agent.

